

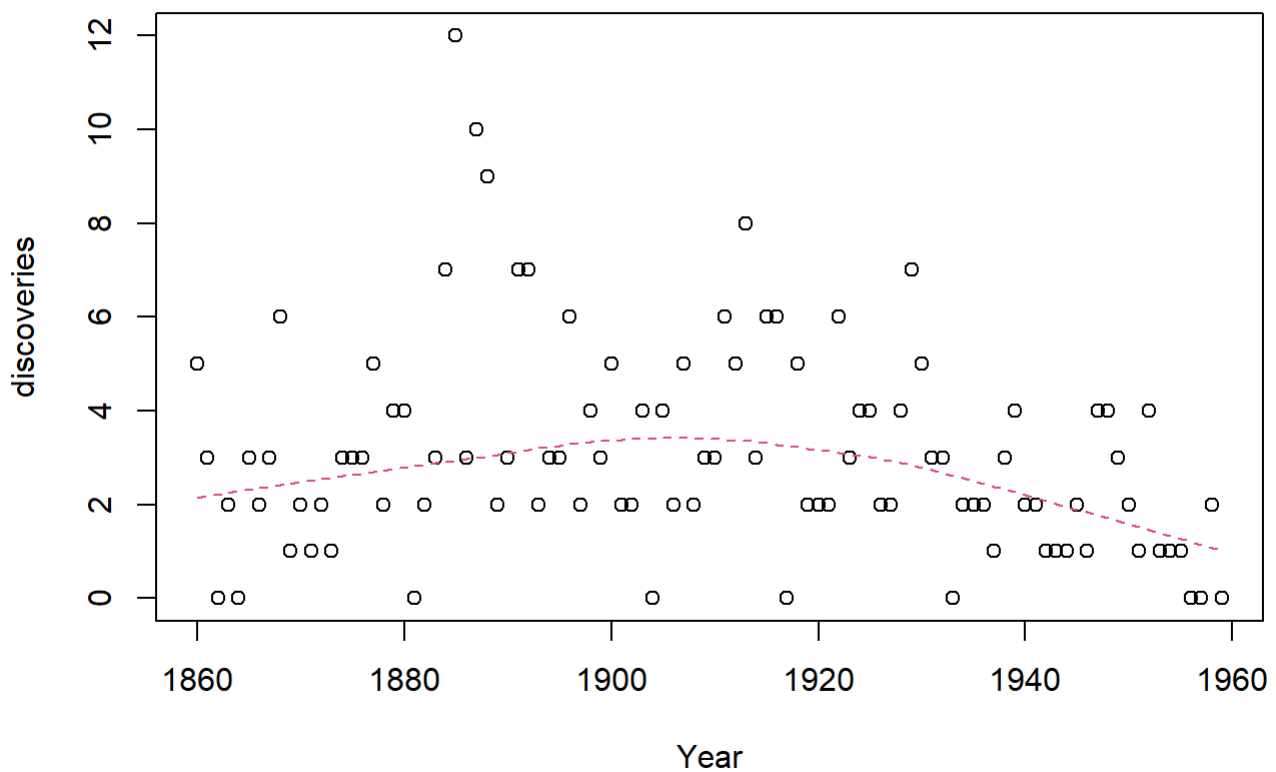
Week11 HW

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1. The dataset discoveries lists the numbers of “great” inventions and scientific discoveries in each year from 1860 to 1959. (a) Plot the discoveries over time and comment on the trend, if any.

```
x<-1860:1959
plot(x,discoveries,xlab="Year")
lines(lowess(x, discoveries, f = 2/3), col = 2, lty=2)
```



From the plot, the number of discoveries appears to increase initially and then decrease over time. The overall pattern suggests a possible quadratic relationship between the number of discoveries and year.

(b) Fit a Poisson response model with a constant term. Now compute the mean number of discoveries per year. What is the relationship between this mean and the coefficient seen in the model?

```
r<-glm(discoveries~1, family="poisson")
summary(r)
```

```
##
## Call:
## glm(formula = discoveries ~ 1, family = "poisson")
##
## Coefficients:
##             Estimate Std. Error z value Pr(>|z|)
## (Intercept)  1.1314      0.0568   19.92  <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for poisson family taken to be 1)
##
##      Null deviance: 164.68  on 99  degrees of freedom
## Residual deviance: 164.68  on 99  degrees of freedom
## AIC: 435.69
##
## Number of Fisher Scoring iterations: 5
```

```
mean(discoveries)
```

```
## [1] 3.1
```

```
exp(r$coefficients)
```

```
## (Intercept)
##           3.1
```

The mean number estimated from Poisson model is identical to the sample mean of the response. In a Poisson model with only an intercept, the maximum likelihood estimator of the mean is the sample mean, and the intercept estimate is the logarithm of this mean.

(c) Use the deviance from the model to check whether the model fits the data. What does this say about whether the rate of discoveries is constant over time?

```
pchisq(deviance(r),df.residual(r),lower=FALSE)
```

```
## [1] 3.79455e-05
```

Under the null hypothesis that the Poisson model with only an intercept is correctly specified, the residual deviance approximately follows a chi-square distribution with $df=99$.

Since the p-value is less than 0.05, we reject the null hypothesis and conclude that the model does not fit the data adequately.

This suggests that the discovery rate is unlikely to be constant over time and that additional explanatory variables, such as year, may be needed.

(d) Fit a Poisson response model that is quadratic in the year. Test for the significance of the quadratic term. What does this say about the presence of a trend in discovery?

```
r1 <- glm(discoveries ~ x, family="poisson")
r2<-glm(discoveries~poly(x,2), family="poisson")
summary(r2)
```

```
##
## Call:
## glm(formula = discoveries ~ poly(x, 2), family = "poisson")
##
## Coefficients:
##             Estimate Std. Error z value Pr(>|z|)
## (Intercept)  1.07207    0.06064  17.681  < 2e-16 ***
## poly(x, 2)1 -2.04766    0.66937  -3.059  0.00222 **
## poly(x, 2)2 -3.05975    0.64821  -4.720  2.35e-06 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for poisson family taken to be 1)
##
##      Null deviance: 164.68  on 99  degrees of freedom
## Residual deviance: 132.84  on 97  degrees of freedom
## AIC: 407.85
##
## Number of Fisher Scoring iterations: 5
```

```
anova(r,r2)
```

```
## Analysis of Deviance Table
##
## Model 1: discoveries ~ 1
## Model 2: discoveries ~ poly(x, 2)
##   Resid. Df Resid. Dev Df Deviance  Pr(>Chi)
## 1          99      164.69
## 2          97      132.84  2   31.846 1.215e-07 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
anova(r1,r2)
```

```
## Analysis of Deviance Table
##
## Model 1: discoveries ~ x
## Model 2: discoveries ~ poly(x, 2)
##   Resid. Df Resid. Dev Df Deviance  Pr(>Chi)
## 1          98      157.32
## 2          97      132.84  1   24.477 7.519e-07 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

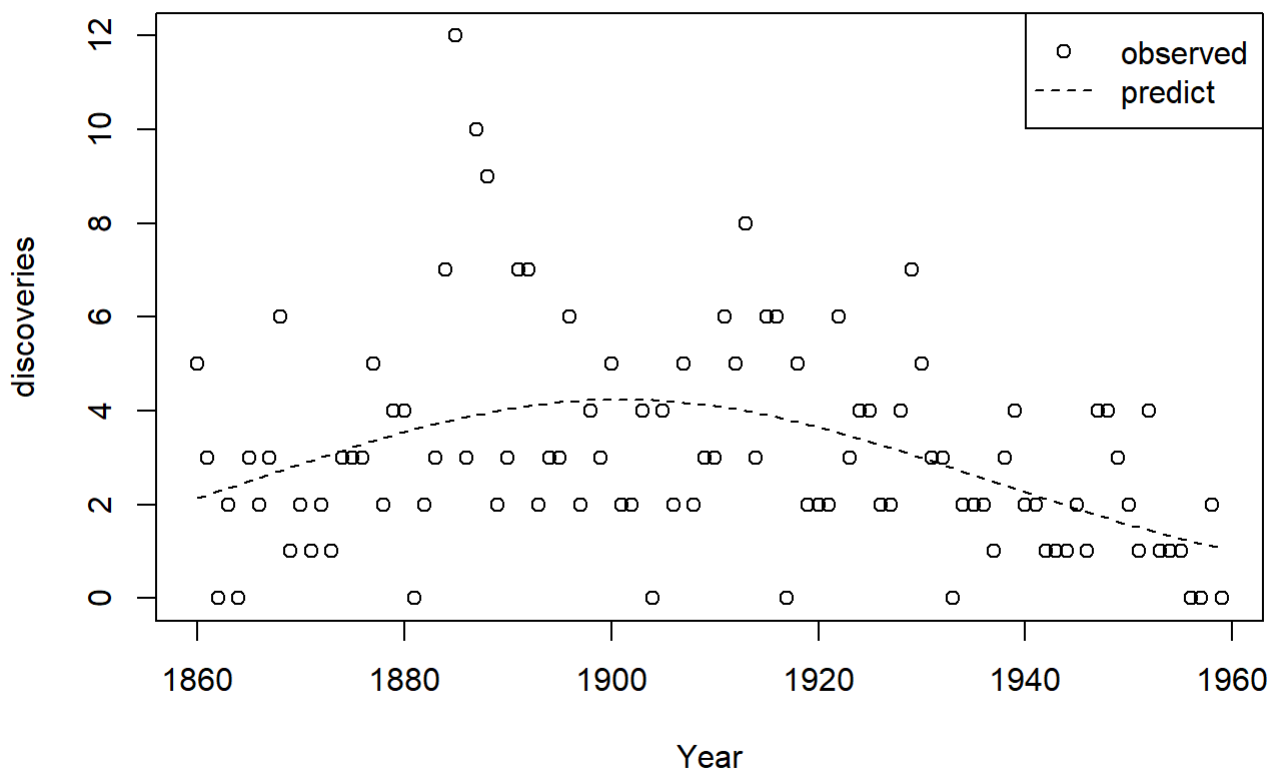
The likelihood ratio test comparing the linear and quadratic models yields a significant p-value. Therefore, the quadratic term is statistically significant.

This provides evidence that the trend in discoveries is not simply linear and that a curved trend is present over time.

(e) Compute the predicted number of discoveries each year and show these predictions as a line drawn over the data. Comment on what you see.

```
y.pred<-predict(r2,type="response")
plot(x,discoveries,xlab="Year")
lines(x,r2$fitted.values,lty=2)

legend("topright",c("observed","predict"),lty=c(NA,2),pch=c(1,NA))
```



The fitted quadratic Poisson model captures the overall long-term trend in the data. The predicted number of discoveries increases during the early years and decreases in later years.

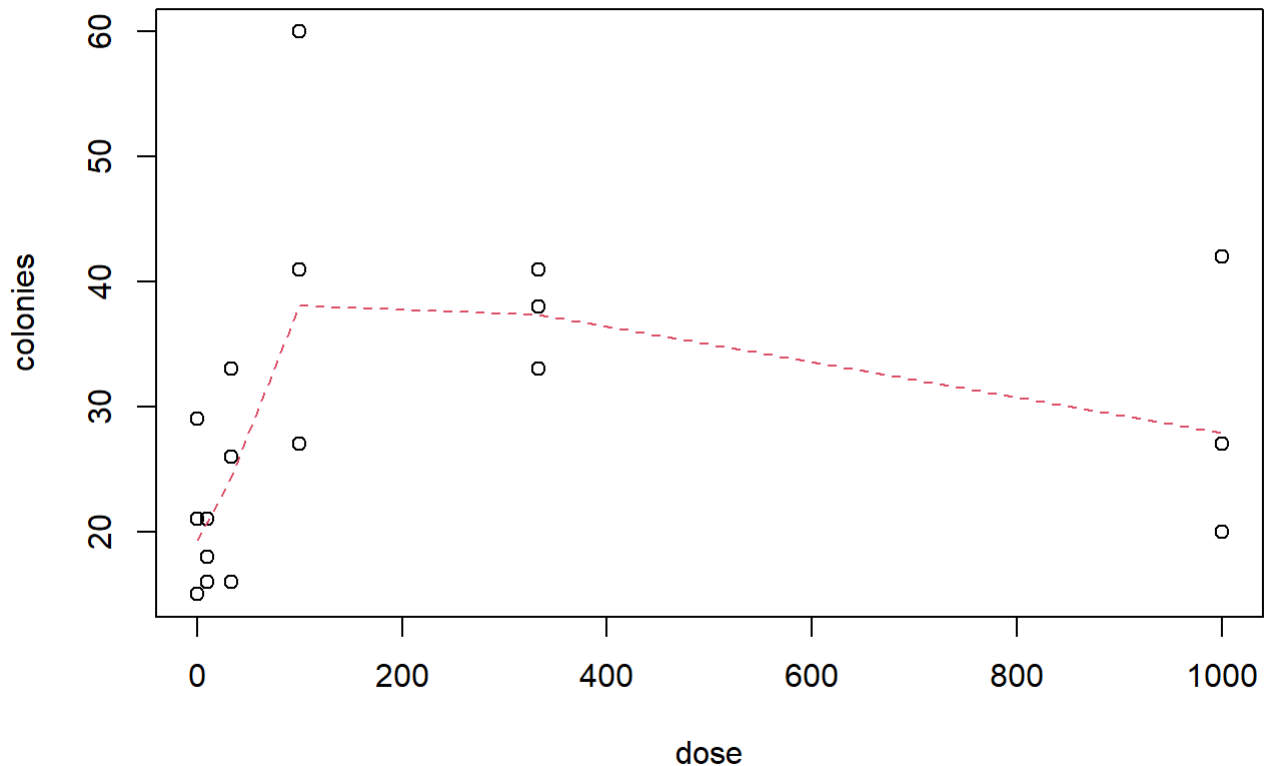
Although substantial year-to-year variation remains, the fitted curve follows the general pattern observed in the data, indicating that the quadratic model provides a much better description than the constant-rate model.

2. The salmonella data was collected in a salmonella reverse mutagenicity assay. The predictor is the dose level of quinoline and the response is the numbers of revertant colonies of TA98 salmonella observed on each of three replicate plates.

(a) Plot the data and comment on the relationship between dose and colonies.

```
library(faraway)
y <- salmonella$colonies
x <- salmonella$dose
plot(x,y,xlab="dose",ylab="colonies")

lines(lowess(x, y, f = 2/3), col = 2, lty=2)
```

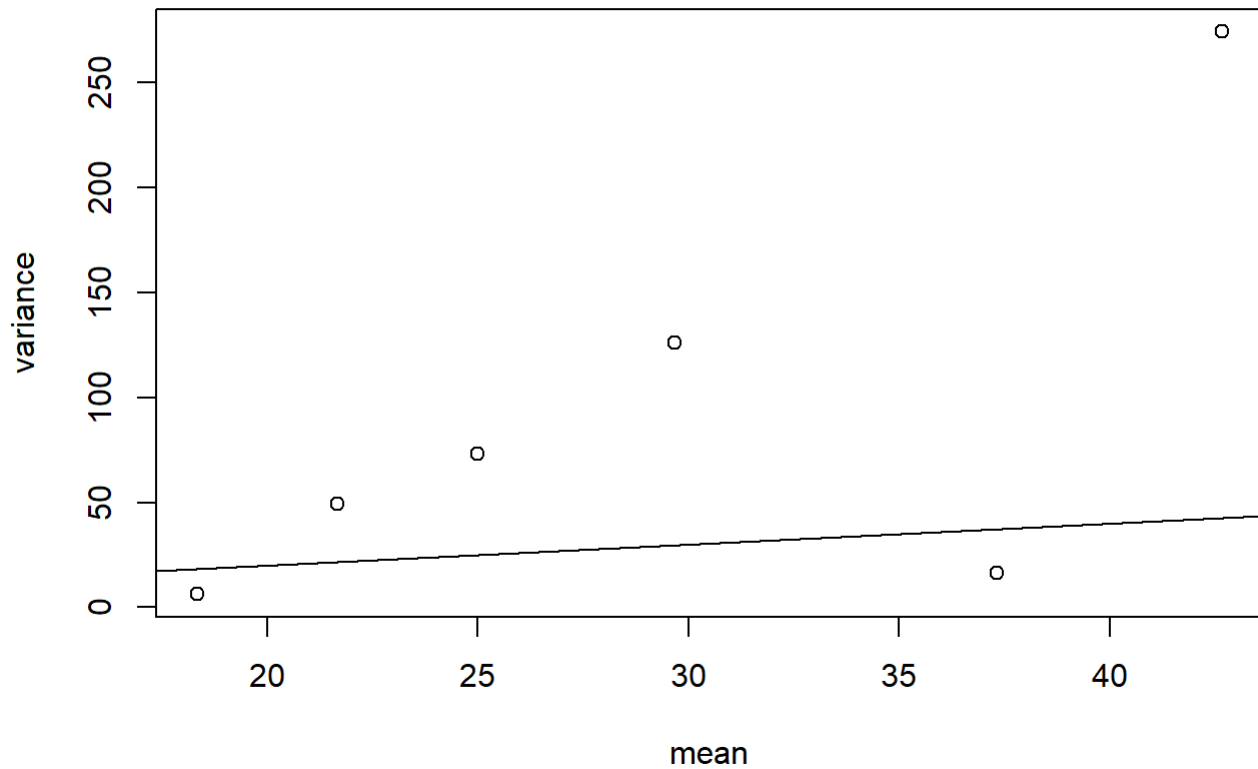


The scatterplot shows that the number of colonies generally increases as the dose increases. However, the relationship does not appear to be linear. The response increases rapidly at moderate doses and then levels off or decreases slightly at higher doses, suggesting a nonlinear dose-response relationship.

(b) Compute the mean and variance within each set of observations with the same dose. Plot the variance against the mean and comment on what this says about overdispersion.

```
m <- tapply(y,x,mean)
v <- tapply(y,x,var)

plot(m,v,xlab="mean",ylab="variance")
abline(a=0,b=1)
```



For a Poisson distribution, the variance should be approximately equal to the mean. The plot shows that most variances are substantially larger than the corresponding means. This indicates the presence of overdispersion, suggesting that a standard Poisson model may underestimate the variability in the data.

(c) Fit a Poisson model with dose treated as a six-level factor. Check the deviance to determine whether this model fits the data.

```
r<-glm(y~as.factor(x), family="poisson")
summary(r)
```

```
##
## Call:
## glm(formula = y ~ as.factor(x), family = "poisson")
##
## Coefficients:
##              Estimate Std. Error z value Pr(>|z|)
## (Intercept)      3.0758     0.1240  24.798 < 2e-16 ***
## as.factor(x)10    -0.1671     0.1832  -0.912 0.361869
## as.factor(x)33     0.1431     0.1695   0.844 0.398427
## as.factor(x)100    0.6776     0.1523   4.449 8.62e-06 ***
## as.factor(x)333    0.5441     0.1559   3.490 0.000484 ***
## as.factor(x)1000   0.3142     0.1632   1.926 0.054099 .
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for poisson family taken to be 1)
##
##      Null deviance: 78.358  on 17  degrees of freedom
## Residual deviance: 33.496  on 12  degrees of freedom
## AIC: 138.03
##
## Number of Fisher Scoring iterations: 4
```

```
pchisq(deviance(r),df.residual(r),lower=FALSE)
```

```
## [1] 0.0008096273
```

The residual deviance is significantly larger than expected under the Poisson model. Therefore, we reject the null hypothesis that the model provides an adequate fit.

The lack of fit may be caused by overdispersion.

(d) Fit a Poisson model what includes an overdispersion parameter. Describe the key differences in the results between this model and the one fitted in part (c).

```
(dp <- sum(residuals(r,type="pearson")^2)/r$df.res)
```

```
## [1] 2.777999
```

```
summary(r,dispersion=dp)
```

```
##
## Call:
## glm(formula = y ~ as.factor(x), family = "poisson")
##
## Coefficients:
##              Estimate Std. Error z value Pr(>|z|)
## (Intercept)      3.0758      0.2067  14.878  <2e-16 ***
## as.factor(x)10    -0.1671      0.3054  -0.547   0.5843
## as.factor(x)33     0.1431      0.2825   0.507   0.6124
## as.factor(x)100    0.6776      0.2539   2.669   0.0076 **
## as.factor(x)333    0.5441      0.2599   2.094   0.0363 *
## as.factor(x)1000   0.3142      0.2719   1.156   0.2479
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for poisson family taken to be 2.777999)
##
##      Null deviance: 78.358  on 17  degrees of freedom
## Residual deviance: 33.496  on 12  degrees of freedom
## AIC: 138.03
##
## Number of Fisher Scoring iterations: 4
```

The estimated coefficients are identical to those from the Poisson model because both models have the same mean structure.

However, the standard errors are larger under the quasi-Poisson model because the extra variability is taken into account.

Consequently, the test statistics are smaller and the p-values are larger.

(e) Give the predicted mean response for a dose of 500. Compute a 95% confidence interval.

Since a prediction at dose 500 is required and 500 is not one of the observed dose levels, we fit a quadratic quasi-Poisson model with dose treated as a continuous predictor.

```
r2<-glm(y~x + I(x^2), family="quasipoisson")

(dp <- sum(residuals(r2,type="pearson")^2)/r2$df.res)
```

```
## [1] 4.126221
```

```
p <- predict(r2, newdata=data.frame(x=500), se.fit=TRUE)

eta <- p$fit
exp(eta)
```

```
##      1
## 46.66347
```

```
se <- p$se.fit

(ci <- exp(eta+c(-1,1)*1.96*se))
```



```
## [1] 31.14967 69.90377
```

The estimated mean response at a dose of 500 is 46.66 colonies. The corresponding 95% confidence interval is (31.15, 69.90).

(f) At what dose does the maximum predicted response occur?

```
b <- coef(r2)
(x.max <- -b[2]/(2*b[3]))
```

```
##          x
## 546.4212
```

The fitted quadratic model implies that the predicted response reaches its maximum at 546.42.

(g) Fit a Negative Binomial model to the data. Use this model to repeat the analysis required in parts (e) and (f).

```
library(MASS)

r.nb <- glm.nb(y ~ x + I(x^2))
#summary(r.nb)

p.nb <- predict(r.nb,
                newdata=data.frame(x=500),
                se.fit=TRUE)

eta <- p.nb$fit
se <- p.nb$se.fit

mu.nb <- exp(eta)

ci.nb <- exp(eta + c(-1,1)*1.96*se)

mu.nb
```

```
##          1
## 48.85571
```

```
ci.nb
```

```
## [1] 33.20539 71.88233
```

Under the Negative Binomial model, the predicted mean response at dose 500 is 48.86 colonies, with a 95% confidence interval of (33.21, 71.88). The estimated mean is slightly larger than that obtained from the Poisson model (46.66 colonies).

```
b <- coef(r.nb)

x.max.nb <- -b[2]/(2*b[3])

x.max.nb
```

```
##          x
## 546.3227
```

The maximum predicted response under the Negative Binomial model occurs at a dose of approximately 546.32, which is very similar to the value obtained under the quasi-Poisson model (546.42).